GeOSM

Complete user guide

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1. Map and navigation

Interactive map view

GeOSM displays an interactive map built on OpenLayers, with smooth navigation via mouse, trackpad or touch (zoom, pan, rotate). The current position and zoom level are preserved when switching tools, so you never lose track of what you were exploring.

> **Example :** You explore the Bastos neighborhood in Yaoundé by progressively zooming in from the country view down to street level, without ever reloading the page.

Navigation history

Just like a web browser, GeOSM keeps a history of visited positions and zoom levels. The toolbar's back and forward buttons let you quickly return to a previously explored view without having to find it again manually.

> **Example :** After zooming into three different cities to compare their school infrastructure, you return to the first view in one click using the back button.

Context menu (right-click)

Right-clicking (or long-pressing on mobile) anywhere on the map opens a context menu with the actions available at that exact spot: copy coordinates, center the view, start a measurement, create a geosignet, and more.

> **Example :** You right-click on an intersection to instantly copy its GPS coordinates and send them to a colleague in the field.

Zoom to GPS coordinates

If you already know the exact coordinates of a place (for example recorded with a field GPS unit), you can enter them directly so the map centers and zooms in instantly, without going through a name search.

> **Example :** A field agent receives the coordinates "3.8480, 11.5021" by text message and enters them directly to locate the exact point on the map.

Geosignets (saved positions)

A geosignet saves a map position (center and zoom level) under a name of your choice, so you can return to it in one click from any future session, without having to navigate or search for the place again.

> **Example :** You regularly check the Bonapriso neighborhood: you save it as a "Bonapriso" geosignet and find it instantly on every visit.

My Maps (layer compositions)

"My Maps" lets you save a specific combination of active layers (with their opacity and order) under a name, so you can find and reactivate it in one action later, instead of turning on each layer one by one.

> **Example :** You build a "School infrastructure" map combining primary schools, colleges and universities, and recall it in one click at every follow-up meeting.

Map settings

The Settings panel gathers the map's display preferences: interface language (French, English, Spanish), units of measurement, and other options that apply instantly without reloading the page.

> **Example :** An English-speaking user switches the interface to English from the Settings panel in a single action.

Swipe map comparison

The swipe comparison tool overlays two base maps (for example a recent satellite view and an older one) with a slider you drag to progressively reveal one or the other, useful for observing changes in the terrain.

> **Example :** You compare a 2015 satellite image with a 2024 one to visualize urban expansion in a Douala neighborhood.

Descriptive sheet and information popup

Clicking on a map object displays a popup with its main information; a "learn more" button opens an enriched descriptive sheet with links to Wikipedia, Wikidata and the corresponding OpenStreetMap entry when available.

> **Example :** You click on a historic landmark and go straight to its Wikipedia page to learn more about its history.

2. Search and discovery

Geographic search (Nominatim)

The search bar lets you find an address, place name or point of interest by name, powered by the Nominatim geocoding engine. The map automatically centers on the first relevant result.

> **Example :** You type "Marché Central Yaoundé" into the search bar and the map immediately centers on its exact location.

Personalized suggestions

As soon as the map opens, GeOSM proposes a list of suggested layers, computed from the layers most frequently activated by all users of the instance, saving you time browsing the whole catalog.

> **Example :** When opening the Cameroon map, you immediately see "Hospitals", "Schools" and "Main roads" suggested, the most consulted layers.

Hierarchical layer catalog

The catalog organizes all available layers into a Groups > Sub-groups > Layers tree (for example Health > Hospitals), making it easy to navigate even when dozens of layers are available.

> **Example :** You're looking for transport data: you expand the "Transport" group then the "Roads" sub-group to find the exact layer you need.

Active layers (opacity, heatmap, resync)

Once a layer is activated, a dedicated panel lets you adjust its opacity, switch to a heatmap representation to visualize concentrations, and resync it if the source data has been updated on the admin side.

> **Example :** You lower the opacity of the "Administrative boundaries" layer to 40% to better see the satellite base underneath, then switch the "Hospitals" layer to heatmap view to spot underserved areas.

Base map switching

The base map selector lets you switch between several base styles (standard, satellite, terrain...) without losing active layers or the current position.

> **Example :** You switch from the "Standard" base map to "Satellite" to visually check whether a drawn road matches an existing track.

Legend

The legend shows, for each active layer, the meaning of the colors, symbols and icons used, updating automatically as you activate or deactivate layers.

> **Example :** After activating the "Land use" layer, you check the legend to understand what each shade of green represents.

Layer recommendations (co-activation)

In addition to the initial suggestions, GeOSM proposes complementary layers to the ones already active, based on layer combinations frequently activated together by other users.

> **Example :** You activate the "Hospitals" layer and GeOSM suggests "Pharmacies" and "Health centers", often consulted alongside it.

3. Analysis and production tools

Drawing (points, lines, polygons)

The drawing tool lets you sketch points, lines and polygons directly on the map, with the drawing saved so you can find it later or share it with other users of the instance.

> **Example :** A municipal official draws the outlines of flood-prone areas in his neighborhood to share them with his technical team.

Measurement (distance and area)

The measurement tool computes, in real time, the distance of a path or the area of a polygon drawn directly on the map, without needing an external GIS tool for a quick estimate.

> **Example :** You trace the boundary of a plot of land to quickly estimate its area before a field visit.

Routing (OSRM)

Route calculation relies on the OSRM engine to propose a real road route between two points, with distance, estimated duration and the path displayed directly on the map.

> **Example :** You trace a route from Douala to Yaoundé and get a distance of 240 km, an estimated duration and the full path on the map.

Nearest search by real road distance

Rather than relying on straight-line distance, this search finds the nearest entity (a hospital, a school...) accounting for the actual road network, avoiding errors when two geometrically close places are actually separated by an obstacle.

> **Example :** You click on the map to find out which hospital is actually closest by travel time, when two hospitals seem equidistant as the crow flies but one is separated by a river with no bridge.

Spatial analysis (buffer, intersection, union, difference)

The spatial analysis module lets you generate a buffer zone around an object, or combine several geometries via intersection, union or difference, to answer land-use planning questions.

> **Example :** An urban planner creates a 500 m buffer zone around a school to identify all buildings in its immediate vicinity.

Multi-format export

Layer data can be exported in several standard formats (including Shapefile and GeoJSON) for reuse in a desktop GIS tool like QGIS.

> **Example :** A researcher downloads Cameroon's school data in Shapefile format to analyze it in QGIS Desktop.

Printing

The print function generates a printable version of the current map view, with active layers, for offline use or inclusion in a report.

> **Example :** You print the current view showing a neighborhood's schools for inclusion in a paper report for a community meeting.

PDF location plan

The location plan generates a professionally laid-out PDF document (A4 format) centered on a chosen point, with a title, a landmark and the surrounding cartographic context, ideal for reaching a specific site in the field.

> **Example :** A field agent picks a point on the map, enters a title and a landmark, then generates an A4 location plan to print for reaching the site.

Geolocated comments

Comments let you pin a remark, a report or a question directly to a point on the map; each pin opens a full discussion thread, with the ability to reply and mark the topic as resolved.

> **Example :** An editor replies to a pothole report to say that repairs are underway, then marks the comment as resolved once the work is done.

Elevation

Clicking on the map shows the elevation of the selected point (from SRTM data); for a traced path, a full elevation profile shows the grade changes along the route.

> **Example :** A road engineer traces a planned road and views the elevation changes along the route to anticipate steep sections.

Mapillary (street-level imagery)

The Mapillary integration displays, where available, street-level panoramic photos at the selected map location, to get a concrete view of a place without traveling there.

> **Example :** Before heading to a site, you check the Mapillary view of an intersection to spot the surrounding shops.

Statistics (including AI-narrated statistics)

Every layer has a statistics panel (entity count, distribution...); the AI assistant can also turn these figures into a clear narrative sentence rather than a plain raw table.

> **Example :** You activate the "Hospitals" layer and see: "This layer counts 342 hospitals spread across the territory, with a stronger concentration around Yaoundé and Douala."

4. AI Assistant

Natural language queries

The conversational assistant understands questions asked in natural language (in both French and English) about the map's data and features, without requiring you to know any particular syntax.

> **Example :** You type "How many hospitals are there around Yaoundé?" and get a direct answer in natural language.

Assistant-driven map actions

Beyond answering, the assistant can act directly on the map (activate a layer, zoom to an area, run a spatial analysis) thanks to Gemini function-calling, turning a conversation into a real remote control for the map.

> **Example :** You type "Show me the hospitals within 5 km of downtown and zoom in"; the assistant activates the Hospitals layer, runs a proximity analysis, and automatically zooms to the result.

Location plans drafted via chat

From the conversation with the assistant, you can request the creation of a location plan; the AI can also automatically draft the description and landmark when you only provide the title and coordinates.

> **Example :** A field agent in a hurry enters just a title and coordinates, checks "Auto-draft", and lets the AI propose a description and landmark that he corrects before submitting.

Narrated summaries and statistics

The assistant can summarize the current map view or a layer's statistics in a single sentence, handy for a quick report without having to interpret raw figures yourself.

> **Example :** You click "Summarize view" after navigating the map and get a paragraph describing what you're looking at, useful for a quick report or an email.

Conversation history

Every conversation with the assistant is kept and linked to your user account, so you can resume a previous exchange or refer back to it later without retyping everything.

> **Example :** A week later, you find the exchange in which the assistant helped you locate schools in a specific neighborhood.

5. Sharing

Social media sharing

A share button lets you publish the current map view directly to social media, to quickly spread geographic information to a wide audience.

> **Example :** You share a view showing the progress of new road works on social media.

Read-only shared maps

A share link gives access to a specific map view (active layers, position) in read-only mode, viewable by anyone without needing to create an account.

> **Example :** You share a view showing Yaoundé's schools and hospitals with a colleague via a short link, which he views without needing to log in.

6. Account management

Classic login (email/password)

Email and password login remains available to every user, with email verification at sign-up and a password reset procedure in case of a forgotten password.

> **Example :** You log in with your work email and password to access editing features reserved for editors.

One-click login via OpenStreetMap (OAuth 2.0)

An OpenStreetMap contributor already used to their osm.org account can log in with one click via "Log in with OpenStreetMap", without creating a new password; a local GeOSM account is automatically created on first login.

> **Example :** An OSM contributor clicks "Log in with OpenStreetMap" and immediately accesses the geoportal without going through a classic sign-up.

Profile (including linked OpenStreetMap profile)

The profile panel shows your personal information and, if an OpenStreetMap account is linked, its display name, creation date and number of contributions, with the ability to link or unlink this account at any time.

> **Example :** You check your profile to see how long your OpenStreetMap account has existed and how many contributions are linked to it.

7. Administration

Instance (country) management

Restricted to authorized accounts, instance management lets you create, configure and administer a GeOSM deployment for a given country, with its own layers, users and themes.

> **Example :** The super administrator deploys GeOSM for a new African country, Togo, by defining its coordinates and instance settings.

Theme and layer management

Administrators and editors can create and organize themes (groups and sub-groups), as well as create, style and update the data layers attached to them.

> **Example :** The editor creates a "Primary schools" point layer, attached to the "Primary education" sub-group.

User and role management

Authorized accounts can create user accounts without going through public sign-up, and assign the appropriate role (Super Administrator, Instance Administrator, Editor, Viewer) according to each person's responsibilities.

> **Example :** The super administrator creates an account for a new instance manager and assigns them the Instance Administrator role.